

Instructions

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Your Task

In this lab, you will work with MPI derived datatypes to send and receive a user-defined structure between MPI processes. This assignment will help you understand how to represent complex C++ structures as MPI datatypes using `MPI_Type_create_struct`.

Objective

You are provided with a partial C++ program that defines a `Point3D` structure representing a point in 3D space, with three double precision values: `x`, `y`, and `z`. Your task is to create and use an MPI derived datatype that matches this structure, and use it to send and receive the data between two processes.

Implementation Instructions

The core structure of the program has been provided. Your job is to complete the MPI sections in the TODO comments:

- 1. **Define a derived MPI datatype** for the `Point3D` structure using `MPI_Type_create_struct`.
- 2. **Commit the datatype** using `MPI_Type_commit` so it can be used in communication.
- 3. **Send the Point3D structure** from one MPI process to another using `MPI_Send`.
- 4. **Receive the Point3D structure** on the receiving process using `MPI_Recv`.

The rest of the program has been provided. You only need to implement the MPI datatype definition and communication calls.

Compilation and Execution

Submission

Submit the following file:

- `mpi_derived_type.cpp`: Your completed C++ source code with the MPI datatype implementation.

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